



GC1015

DATA QUALITY, DISTRIBUTIONS, AND DESCRIPTIVE EVIDENCE

Session 3 connects trustworthy data preparation to defensible summaries, honest visuals, and business decisions.



Use what you already know

RECALL

Variables, measurement scales and grain

1

Recall the language

- An observation is one recorded item or event; a variable is one recorded attribute. Why is an order identification number classified as nominal (an unordered category) despite containing only digits?

2

Connect to today

- **Grain** means what one row represents. Measurement scales describe what values mean and which calculations fit. Today we use these ideas to check records and handle missing values.

CHOOSE, THEN EXPLAIN

Why is an order identification number classified as nominal categorical despite containing only digits?

- A** The digits identify an order; they do not measure an amount
- B** An order numbered 200 represents twice as much as order 100
- C** Averaging order numbers tells us how many orders were placed

Choose first. Compare your reason with a neighbor.



Use what you already know

RECALL

Key performance indicators, mean and median

1

Name the business measure

- A key performance indicator (**KPI**) is a measure used to track a business goal, such as delivery reliability.

2

Recall and connect

- The **mean** is the arithmetic average; the **median** is the middle value after sorting. Why should an analyst examine both for late delivery minutes?
- A resistant summary changes less when a few values are unusually large or small. Today we compare summaries and explain what each tells us.

CHOOSE, THEN EXPLAIN

Why should an analyst examine both the mean and median for late delivery minutes?

- ☐ A Their difference can reveal the influence of unusually long delays
- ☐ B The mean is unaffected by unusually long delays
- ☐ C The mean and median always give the same result

Choose first. Compare your reason with a neighbor.



Data quality · Distributions
· Visual evidence ·
Comparison

ONE INVESTIGATION, FOUR STEPS

Data quality · Distributions · Visual evidence · Comparison

1

First, trust the inputs

- Check what each row represents, which values are missing, and which calculations fit the variable.

2

Then, compare the evidence

- **A distribution is the pattern of values:** their center (typical value), spread (variation) and shape. Compare these features across hubs, the delivery centers.
- Each section gives you another reason to keep or revise your decision.



Slides hand off to the lab,
the lab hands back

TODAY'S LOOP: SLIDES AND LAB

Use the lab guide and workbook, then bring results back to class

1

Sections before the break

- At each Practice slide, open the matching Part in the lab guide and workbook. Complete the Excel cells shown, check the feedback, and save.

2

Your second analysis tool

- **SPSS** originally meant Statistical Package for the Social Sciences. We use **IBM SPSS** Statistics to check summaries and charts. Supplied syntax means saved software commands.

3

Discuss and revisit

- After the break, finish Sections 3 and 4. The evidence clinic is time to explain results and compare Excel with **SPSS**. Then discuss the two group questions.



Why it matters

WHY THIS INVESTIGATION MATTERS

Better evidence supports better decisions

1

A decision affects people

- A penalty is a financial charge. A carrier is the delivery company. A penalty may encourage responsibility, but a weak comparison can assign blame unfairly. Delaying action also has a cost.

2

Your evidence has a job

- **Use each calculation to answer a question:** what is typical, how variable is service, and what remains unexplained?

DECISION POINT: ACT NOW OR INVESTIGATE?

An early penalty and a delayed response both have costs.



Situation

Metro coordinates deliveries through three hubs. Hub C's customers report delays. Its carrier contract includes a service-level agreement (SLA), the promised service standard. Management can apply a penalty now or investigate until the next scheduled review. The cause is uncertain; Metro must support customers either way.

A

Apply the penalty now

Benefit: accountability now. Risk: penalizing the wrong cause.

B

Hold one cycle to check the records

Benefit: better evidence. Cost: Metro funds support while waiting.

Your decision

Choose A or B. Record your reason; revisit it at the end.

SECTION

DATA QUALITY, PROVENANCE, AND MEASUREMENT SCALES

Provenance means where data came from and how it changed. A sentinel is a special code for a missing value. Check these before deciding which records count.

Field Meanings · Missing Values · Valid Counts

Justice Department orders election-record preservation while seeking voter data

WHAT HAPPENED

The U.S. Justice Department asked 29 states and Washington, D.C., to preserve 2024 election records while seeking detailed voter files. Those files include addresses, birth dates, driver's-license numbers and partial Social Security numbers. The preservation letters cover both paper documents and electronic information.

Today's data-quality work asks when records support a fair comparison. Keep that question in mind as information moves between organizations.

Before combining these voter files, what should an analyst check first?

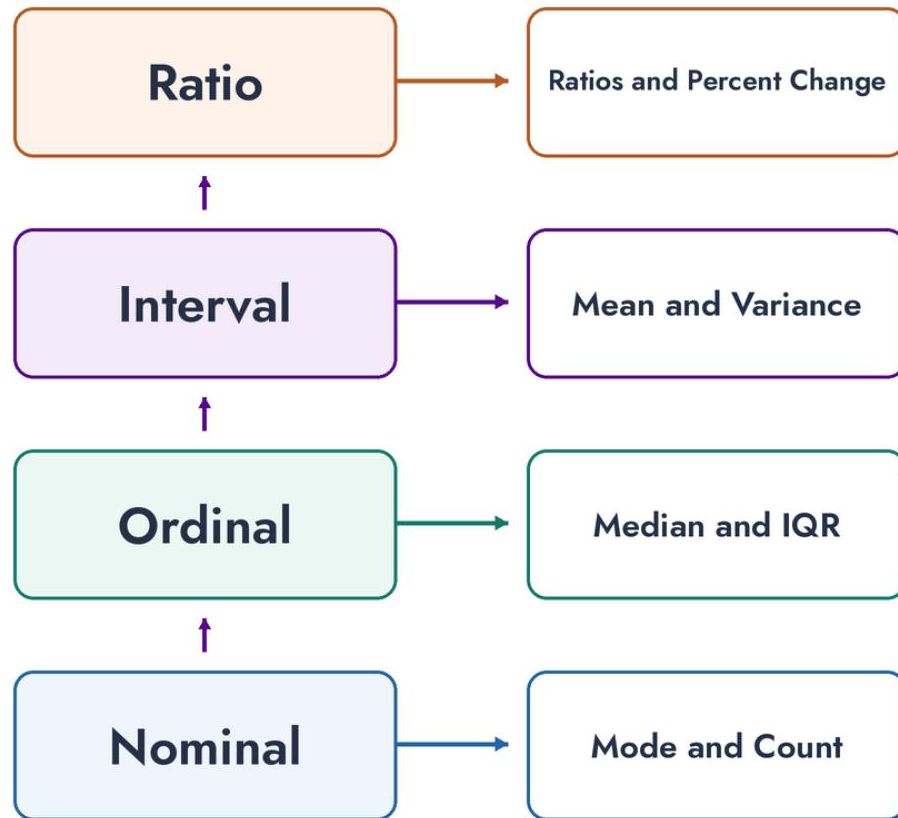
- A** What each field means and how it was recorded
- B** Whether the column names and formats match
- C** Whether the files contain the same row count

Source: The Associated Press · 2026-09-10



Field reporter covering the election-records story. [Illustrative image]

THE MEANING OF MEASUREMENT SCALES IN BUSINESS ANALYTICS



Labels and ranks

Nominal: unordered labels, such as order identifiers (IDs). Ordinal: ranked categories, such as OrderTier 1–3. Ranks do not imply equal gaps.

Differences and ratios

Interval: equal gaps, arbitrary zero (Celsius). Ratio: equal gaps, meaningful zero (elapsed minutes), so 20 minutes is twice 10.

Read the summary names

Mode: most frequent value. Interquartile range (IQR): middle-half width. Variance: spread using squared distances.

Percent change: change relative to the starting value. Choose a summary that fits the scale.

MATHEMATICAL ADMISSIBILITY BY MEASUREMENT SCALE

Scale Hierarchy: $\text{Nominal} \subset \text{Ordinal} \subset \text{Interval} \subset \text{Ratio}$

Mathematical admissibility means choosing calculations that have a useful meaning.

Nominal: unordered labels; count categories. Ordinal: ranked categories; compare order, but do not assume equal gaps.

Interval: equal differences, with an arbitrary zero, such as Celsius temperature. Ratio: equal differences and a meaningful zero, such as elapsed minutes; ratios are meaningful too.

The inclusion signs ($\subset$) indicate added mathematical properties. Numeric storage alone does not establish a scale.

Use the data dictionary, the guide to each field's meaning, units and codes.

SPSS AND EXCEL SCALE VALIDATION SYNTAX

```
* SPSS Syntax for Variable Level Specification and Frequency Screening.  
VARIABLE LEVEL CustomerID (NOMINAL) OrderTier (ORDINAL) NetInvoice (SCALE).  
FREQUENCIES VARIABLES=OrderTier NetInvoice  
  /STATISTICS=STDDEV MINIMUM MAXIMUM MEAN MEDIAN MODE  
  /ORDER=ANALYSIS.
```

Syntax means saved software commands. VARIABLE LEVEL records whether values are nominal labels, ordinal ranks or scale measurements. FREQUENCIES counts occurrences.

A frequency is how often a value occurs. STDDEV requests standard deviation, a measure of spread. We interpret the statistical summaries in the next section.

A metadata setting is a description of the field. It cannot establish that an average has a useful meaning.

This is a supplied reference. Run the delivered LAB01 syntax for the case.



Match meaning to
calculation

WHAT CAN WE CALCULATE?

Match the calculation to the meaning of the variable

1

Check the scale

- Mathematical admissibility means choosing a valid calculation: **nominal** allows counts; **ordinal** adds ranking and medians; **interval** adds means; **ratio** permits ratios.

2

Try two comparisons

- Order tiers 1, 2 and 3 show rank, not equal service gaps. Tier 3 is not three times the service of Tier 1.
- A 20-minute duration is twice a 10-minute duration. A 20°C temperature is not twice as hot as 10°C because Celsius has an arbitrary zero.

3

Use the meaning, not the format

- Digits in a spreadsheet can label a category or measure a quantity. Read the field definition before selecting a summary.



One row, one stated
observation

WHAT DOES ONE ROW REPRESENT?

Check the unit of observation before counting or averaging

1

Name the grain

- Data screening checks **grain**, entry errors and missing values. **Grain** means what one row represents, such as one order or one item within an order.

2

See the counting problem

- One order containing three items may occupy three rows. Counting rows would report three items, not three orders.
- **Conflating** means mixing unlike units. Adding item rows and order rows can double-count activity and distort an average.

3

Check before calculating

- Check the key (identifier expected to distinguish rows) for duplicates (repeated records). Then check each field. Missing age does not invalidate a separately recorded delivery duration.

EXCEL ROBUST DATA SCREENING AND MISSING VALUE RECODING FORMULA ARCHITECTURE

```
=IF(ISBLANK(A2), "MISSING_NULL", IF(A2=-99, "SENTINEL_NULL", IF(A2=999,  
"SENTINEL_NULL", IF(ISNUMBER(A2), IF(AND(A2>=18, A2<=100), A2, "OUT_OF_BOUNDS"),  
"NON_NUMERIC_TEXT"))))
```

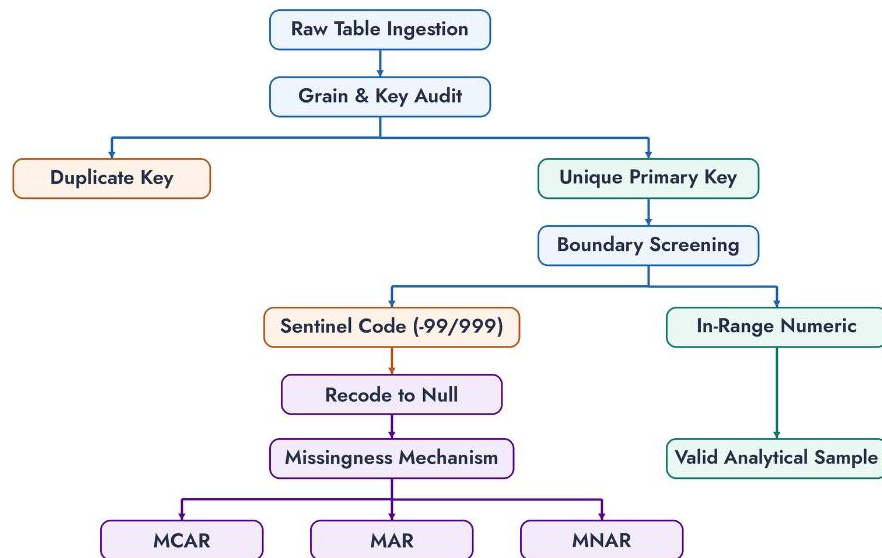
This separate age-screening example uses nested IF: one conditional test inside another. Read from the outside inward: blank, missing code, numeric value, then allowed range.

Null means missing. A matched test returns a label; a valid age passes through unchanged.

This example uses -99. LAB01 uses -1 for missing age and 999 for missing latency, the recorded delivery delay in minutes.

Keep valid latency even when age is missing. Use each variable's own code rule.

SCREENING RECORDS AND UNDERSTANDING MISSINGNESS



Read the workflow and MCAR

Ingestion means importing records. An audit checks records; boundary screening checks allowed values. Null means missing, not zero.

MCAR (Missing Completely at Random): missingness is unrelated to observed or unobserved values.

MAR: Missing at Random

Missingness may depend on recorded information, but not on the missing value after accounting for that information.

MNAR: Missing Not at Random

Missingness still depends on unobserved information. The missing value itself may affect whether it is reported. These are assumptions to investigate; a blank alone cannot identify its cause.

AI SEARCH & SHARE



Artificial intelligence (AI)
tools can help; check their
answers.

APPLY THE IDEA

Artificial intelligence (AI) tools can help; check their answers.

1

Consider

- Why does automated linking in Excel sometimes convert blank source cells into numeric zeros, and how does this distort data screening?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: CAN WE TRUST THE RECORDS?

Keep A or B, or change your mind.



New evidence

The file uses -1 for missing age and 999 for missing latency. A missing age can coexist with a valid delivery time. OrderTier codes rank service levels; their gaps are not measured.

A

Apply the penalty now

Benefit: immediate accountability.

Risk: blaming the carrier for the wrong cause.

B

Hold one cycle for an audit

Benefit: stronger evidence of the cause.

Cost: Metro funds support during the delay.

Your decision

Which records and measures should influence your decision?



Excel first, SPSS second

PRACTICE A: CALCULATE, CHECK, SAVE

Follow the same Part in the lab guide

1

Open

- IN-CLASS_Lab Part A in your saved workbook. The lab guide keeps the formula and explanation together.

2

Do now

- **6 minutes:** complete **B7:B9**. Use the feedback beside the row, then save.

3

Next

- **H1** names the first analysis block in the supplied **SPSS** file. Interpret the results and run **H1** during the Section 4 evidence clinic.

SECTION

DESCRIBING DISTRIBUTIONS: CENTER, SPREAD, AND SHAPE

Compare center (typical value), spread (variation) and shape (the pattern of values). Learn why unusually large or small values affect summaries differently.

August inflation accelerates beneath an unchanged annual headline rate

WHAT HAPPENED

The Consumer Price Index (CPI) tracks U.S. consumer prices. In August, prices were 3.4% above a year earlier, matching July's annual rate. Yet the monthly increase rose from 0.1% to 0.4%. Gasoline rose 3.9% that month, while grocery prices were unchanged.

Managers often receive several summaries of the same data. Before recommending action, consider what each number helps you understand and what it leaves open.

Why can the annual rate stay steady while the monthly increase becomes larger?

- A** They compare prices over different time periods
- B** The monthly calculation must contain an error
- C** Every product must have risen by the same amount

Source: The Associated Press · 2026-09-11



Newsroom reporter covering the inflation story. [Illustrative image]



Three ways to summarize
center

WHAT MEASURES OF CENTER MEAN

Three summaries answer different questions

1

Mean: use every value

- The **mean** is the sum divided by the valid **count**. For 10, 10 and 40 minutes: $60 \div 3 = 20$ minutes.

2

Median and mode

- The **median is the middle ordered value**: 10 minutes. The **mode** is the most frequent value: also 10. The 40-minute delay pulls the **mean** upward.

3

Read Metro's evidence

- Metro's valid latencies have **mean** 28.4 minutes and **median** 22.0 minutes. The gap suggests upper-tail influence; it does not identify the cause.
- Use the **median** to describe a typical order and the **mean** when every minute contributes to total workload.

MEASURES OF CENTRAL TENDENCY

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \tilde{x} = \begin{cases} x_{(\frac{n+1}{2})} & \text{if } n \text{ is odd} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} & \text{if } n \text{ is even} \end{cases}$$

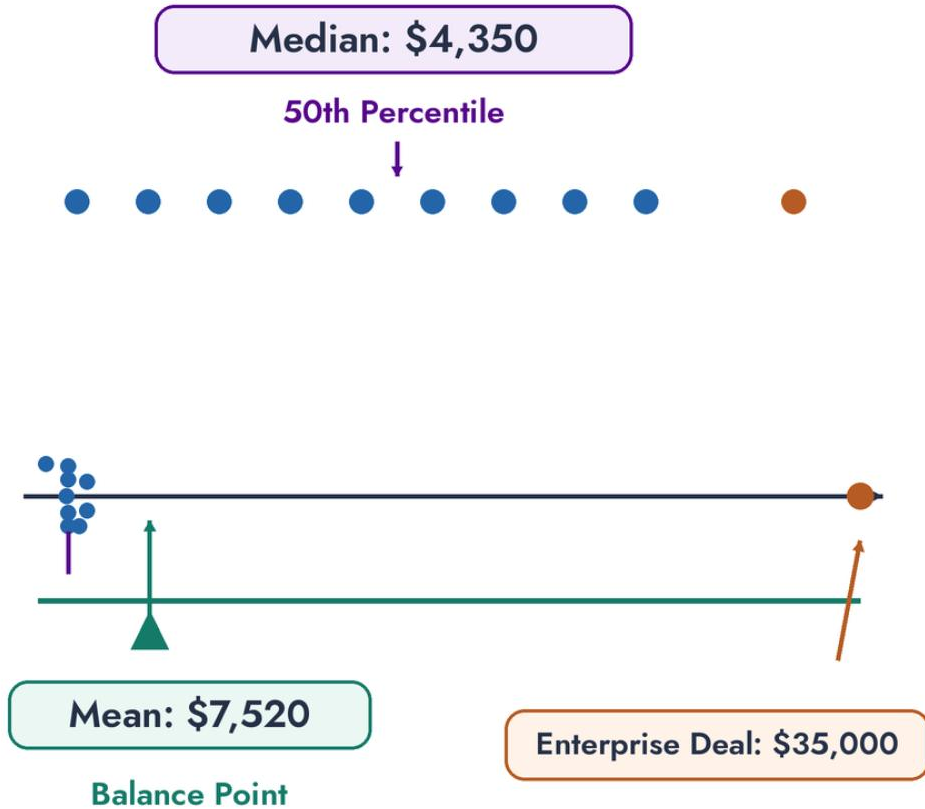
Central tendency means the center or typical value. n is the observation count; x_i is one value; Σ means add the values. $\bar{x}$ is the mean; $\tilde{x}$ is the median. $x_{(k)}$ is the value at sorted position k .

Mean: add, then divide by n . **Median:** sort; find the middle, averaging two middle values when n is even.

For 10, 12, 14, 16, 18 minutes, both are 14. Replace 18 with 118: mean 34, median 14.

The mean uses every magnitude; the median depends on the middle positions.

HOW RESISTANCE AND BREAKDOWN POINTS OPERATE



Resistance: less pull from an extreme

One unusually large sale can pull the mean upward. The median, the middle value, usually moves much less.

Breakdown point: a stress test

This is the share of values that must be replaced to push an estimate without limit. One extreme replacement can break the mean; the median requires about half in large samples.

Keep the business question in view

Resistance is useful for typical experience. A genuine extreme may still matter for cost or risk; investigate it before excluding it.



Read center, spread and
shape

SPREAD AND DISTRIBUTION SHAPE

Center alone does not describe the customer experience

1

Dispersion: how much values vary

- **Dispersion** describes how widely values are spread. **Identical** centers can hide different experiences: 20, 22, 24 and 2, 22, 42 both have **mean** and **median** 22.

2

Skewness: uneven sides

- Skewness describes asymmetry, or unequal shape on the two sides. A right-skewed distribution has a longer upper tail, the region of unusually large values.

3

Kurtosis: the influence of extremes

- **Kurtosis summarizes tail heaviness:** how strongly unusually distant values influence the distribution. Large values can signal more extreme observations. They do not identify an operational cause.

SAMPLE VARIANCE AND SAMPLE STANDARD DEVIATION FORMULA

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}$$

A sample is a subset of the population, the full group of interest. Variance (s^2) summarizes squared deviations from the mean. Standard deviation (SD, s) is its square root in original units.

A deviation is a value minus the mean. x_i is one value; $\bar{x}$ is the mean; n is the count; Σ means add.

For 2, 4, 6 minutes: mean 4; squared deviations 4, 0, 4; variance $8 \div 2 = 4$ square minutes; SD = 2 minutes.

The $n - 1$ divisor reflects degrees of freedom: after estimating the mean, only $n - 1$ deviations vary independently. Extremes strongly influence squared deviations.

EXCEL DESCRIPTIVE DISPERSION AND SHAPE FORMULAS

```
=AVERAGE(B2:B11)
=MEDIAN(B2:B11)
=STDEV.S(B2:B11)
=QUARTILE.EXC(B2:B11, 1)
=QUARTILE.EXC(B2:B11, 3)
=QUARTILE.EXC(B2:B11, 3) – QUARTILE.EXC(B2:B11, 1)
=SKEW(B2:B11)
```

A percentile locates a value in sorted data. Q1 (first quartile) is the 25th percentile; Q3 (third quartile) is the 75th. Interquartile range (IQR) = $Q3 - Q1$, the middle-half width.

AVERAGE and MEDIAN summarize center; STDEV.S gives sample standard deviation. QUARTILE.EXC uses Excel's exclusive percentile method. SKEW and KURT summarize skewness and kurtosis.

B2:B11 means cells B2 through B11 in this separate example. Use the lab's supplied filtered range for its calculations.

EXCEL DESCRIPTIVE DISPERSION AND SHAPE FORMULAS

```
=KURT(B2:B11)
```

A percentile locates a value in sorted data. Q1 (first quartile) is the 25th percentile; Q3 (third quartile) is the 75th. Interquartile range (IQR) = $Q3 - Q1$, the middle-half width.

AVERAGE and MEDIAN summarize center; STDEV.S gives sample standard deviation. QUARTILE.EXC uses Excel's exclusive percentile method. SKEW and KURT summarize skewness and kurtosis.

B2:B11 means cells B2 through B11 in this separate example. Use the lab's supplied filtered range for its calculations.



Keep units and valid
counts visible

HOW SPREAD AND ASYMMETRY ARE CALCULATED

Use the same valid observations for every summary

1

Sample standard deviation

- For Metro's 9,958 valid latencies, sample **SD** is 13.5138 minutes. Subtract the **mean**, square each distance, sum, divide by $n - 1$, then take the square root.

2

Interquartile range

- **IQR** = **Q3** – **Q1** = 31 – 19 = 12 minutes. It describes the width of the middle half, not the full range.

3

Skewness and the shape check

- Cubed distances from the **mean** retain their signs. Positive skewness suggests a longer upper tail.
- Near-zero skewness does not prove a normal, bell-shaped distribution. A histogram groups values into intervals to show their pattern.



Choose a summary for the question

INTERPRETING SPREAD AND SHAPE

Use the summary that fits the service question

1

Compare routine variation and extremes

- **Standard deviation** uses all valid observations and responds strongly to extremes; **IQR** describes the middle half.
- Metro's **SD** is about 13.51 minutes and **IQR** is 12 minutes. These measure different features, so equality is not expected.

2

Choose a service limit with evidence

- Setting a delivery-time limit requires more than center and spread. Inspect the observed tail and the service question before choosing a limit.

3

State the limit of the evidence

- A 22-minute **median** does not promise delivery within 22 minutes. Examine the percentage exceeding a proposed limit and the cost of those delays.

AI SEARCH & SHARE



Discuss with a partner; AI support is optional.

APPLY THE IDEA

Discuss with a partner; AI support is optional.

1

Consider

- Why is kurtosis particularly relevant to financial risk analysts and supply chain planners?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: WHAT COUNTS AS TYPICAL?

Compare the center with the spread.



New evidence

All hubs have a 22-minute median. Hub C has a 29.9-minute mean, versus 27.6 for A/B. Its IQR is 17 minutes, versus 7. Equal typical times can coexist with unequal spread.

A

Apply the penalty now

Benefit: immediate accountability.

Risk: blaming the carrier for the wrong cause.

B

Hold one cycle for an audit

Benefit: stronger evidence of the cause.

Cost: Metro funds support during the delay.

Your decision

Does this evidence change A/B? State what each statistic adds.



Excel first, SPSS second

PRACTICE B: CALCULATE, CHECK, SAVE

Follow the same Part in the lab guide

1

Open

- IN-CLASS_Lab Part B in your saved workbook. The lab guide keeps the formula and explanation together.

2

Do now

- **7 minutes:** complete **B7:B10**. Use the feedback beside the row, then save.

3

Next

- Interpretation and **SPSS H2** belong to the Section 4 evidence clinic.



Pause. Reset. Return ready.



PAUSE AND RESET

Return ready to apply the complete descriptive audit workflow.

Stretch, Hydrate, and Reset

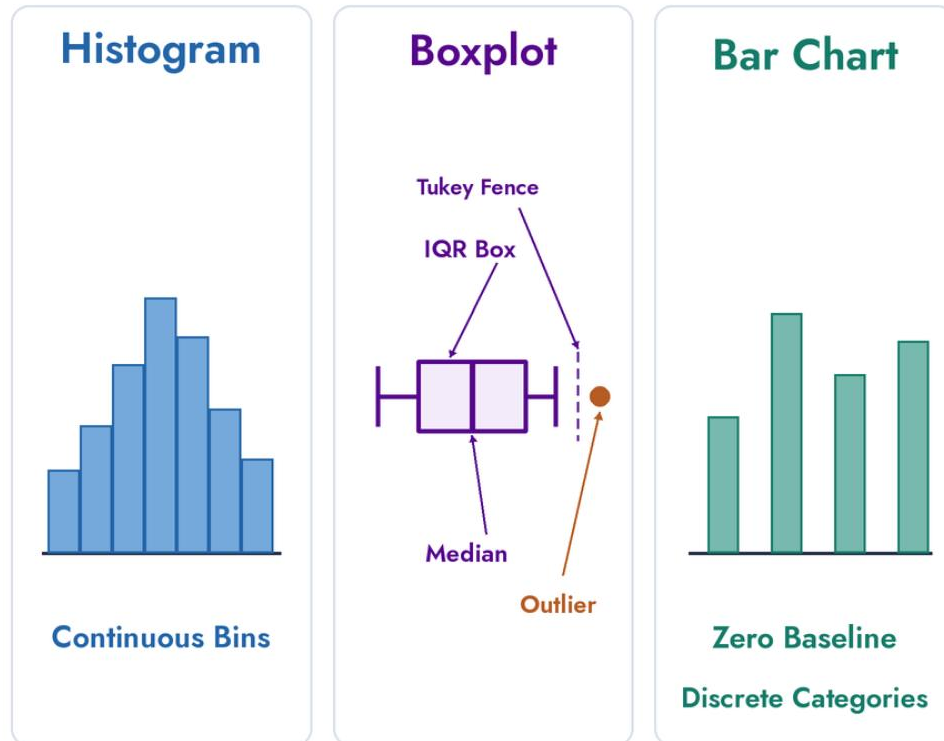
SECTION

VISUAL EVIDENCE, HONEST DISPLAYS, AND DECLUTTERING

Compare bar charts, histograms and boxplots. A boxplot summarizes the middle half and flags unusual values. Remove decoration that makes evidence harder to read.

Chart Choice · Clear Labels · Visual Grouping

VISUAL CHOICE AND DISTRIBUTION GEOMETRY



Bar chart: compare groups

Use separate bars for categories such as hubs. A zero baseline means the value axis starts at zero, so bar lengths remain proportional.

Histogram: see the distribution

Group numeric values into intervals, called bins. Touching bars show how observations are spread across those intervals.

Boxplot: summarize and investigate

The box shows the middle half and its line marks the median. Whiskers are lines extending to observations within the calculated boundaries, called fences. Points beyond them are flags to investigate.

TUKEY OUTLIER FENCE BOUNDARIES

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Lower Fence} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper Fence} = Q_3 + 1.5 \times \text{IQR}$$

$$\text{Extreme Upper Fence} = Q_3 + 3.0 \times \text{IQR}$$

Tukey fences are calculated boundaries for flagging outliers, values unusually far from the others.

$\text{IQR} = Q_3 - Q_1$. Lower fence = $Q_1 - 1.5 \times \text{IQR}$; upper fence = $Q_3 + 1.5 \times \text{IQR}$.

Hub C: $Q_1 = 14$, $Q_3 = 31$, $\text{IQR} = 17$ minutes. Fences are -11.5 and 56.5 minutes. Zero is the physical duration bound, not the calculated lower fence.

Whiskers end at observed values within the fences. A flag starts an investigation; it is not proof of error.

SPSS AND EXCEL FORMULAS FOR DISTRIBUTION ANALYSIS

EXCEL FORMULAS:

<code>=QUARTILE.EXC(C2:C1001, 1)</code>	[Calculate Q1]
<code>=MEDIAN(C2:C1001)</code>	[Calculate Median]
<code>=QUARTILE.EXC(C2:C1001, 3)</code>	[Calculate Q3]
<code>=E2 - D2</code>	[Calculate IQR: Q3 - Q1]
<code>=E2 + 1.5 * F2</code>	[Calculate Upper Fence: Q3 + 1.5*IQR]

Use the same valid durations for Q1, median, Q3 and plots. SPSS HAVERAGE is the named percentile-calculation method used for the lab's numerical comparison with Excel QUARTILE.EXC.

An SPSS EXAMINE boxplot uses Tukey hinges, the lower and upper box edges calculated by a different convention. Points near a boundary may therefore be plotted differently.

Inspect the chart's overall shape. Use the named percentile table for the numerical check, and run LAB01's H3 analysis block rather than typing this reference example.

SPSS AND EXCEL FORMULAS FOR DISTRIBUTION ANALYSIS

SPSS SYNTAX:

```
EXAMINE VARIABLES=Latency_Minutes BY Hub  
  /PLOT BOXPLOT HISTOGRAM  
  /STATISTICS DESCRIPTIVES  
  /NOTOTAL.
```

Use the same valid durations for Q1, median, Q3 and plots. SPSS HAVERAGE is the named percentile-calculation method used for the lab's numerical comparison with Excel QUARTILE.EXC.

An SPSS EXAMINE boxplot uses Tukey hinges, the lower and upper box edges calculated by a different convention. Points near a boundary may therefore be plotted differently.

Inspect the chart's overall shape. Use the named percentile table for the numerical check, and run LAB01's H3 analysis block rather than typing this reference example.



Make comparisons easy to read

CHART GEOMETRY AND BASELINES

Make the picture support an honest comparison

1

Bar charts compare categories

- Use **categorical** gaps between bars and start the value axis at zero when bar length represents amount. A shortened axis can exaggerate a small difference.

2

Histograms show a distribution

- Group numerical observations into intervals, called bins. With equal-width bins, bar height shows **count** or proportion; touching bars indicate adjacent intervals.
- Changing bin width changes the visible detail. State the units and inspect whether the main pattern persists.

3

Boxplots summarize and flag

- **Boxplots** show a **median** and middle-half spread. Points beyond the calculated fences invite investigation. Software quartile conventions can differ; use the lab's specified method.



Keep the evidence easy to find

COGNITIVE LOAD AND VISUAL CLUTTER

Help the reader find the evidence

1

What the reader has to do

- Visual clutter competes for attention. **Cognitive load** is the mental effort needed to interpret the display, including finding labels, comparing values and remembering the question.

2

Simplify without losing meaning

- Removing heavy borders, drop shadows and unnecessary gridlines can make a comparison easier. Keep useful labels, units, source information and meaningful reference lines.

3

Try a reading check

- For a hub comparison, align the bars, use clear hub labels and reserve an accent for the relevant comparison.
- **Then ask:** can a reader identify the measure, largest value and size of the difference without searching through a legend?

DATA-INK RATIO FORMULATION

$$\text{Data-ink ratio} = \frac{\text{ink showing data}}{\text{all ink in the graphic}}$$

Data-ink ratio compares marks carrying data with all marks in a graphic. The numerator (top of the fraction) is data ink; the denominator (bottom) is all ink.

Keep marks that help readers understand values. Remove decoration that competes with them.

Compare the original and simplified charts: can you find the labels, units and important difference more easily?

This is a design principle, not a score to maximize mechanically. Useful axes, context and accessibility may justify additional marks.

DATA-INK RATIO FORMULATION

$$\begin{aligned}\text{Data-ink ratio} &= \frac{\text{data ink}}{\text{all ink}} \\ &= 1 - \frac{\text{non-data ink}}{\text{all ink}}\end{aligned}$$

Data-ink ratio compares marks carrying data with all marks in a graphic. The numerator (top of the fraction) is data ink; the denominator (bottom) is all ink.

Keep marks that help readers understand values. Remove decoration that competes with them.

Compare the original and simplified charts: can you find the labels, units and important difference more easily?

This is a design principle, not a score to maximize mechanically. Useful axes, context and accessibility may justify additional marks.

SPSS GPL DECLUTTERED BAR CHART SPECIFICATION

```
* SPSS GPL Syntax for a Decluttered Horizontal Bar Chart with Monochromatic
Palette.
GGRAPH
  /GRAPHDATASET NAME="graphdataset" VARIABLES=Category Volume MISSING=LISTWISE
REPORTMISSING=NO
  /GRAPHSPEC SOURCE=INLINE.
BEGIN GPL
```

Graphics Production Language (GPL) describes how SPSS draws a graph: its data, axes, labels and marks. Read each block's purpose; you do not need to memorize it.

Check the chart's baseline, units, labels and category order. A neutral palette means restrained colors. A zero baseline preserves bar-length comparisons.

This is optional supplied code. For the required lab, use the delivered syntax and inspect its output. Writing GPL is not an additional task.

SPSS GPL DECLUTTERED BAR CHART SPECIFICATION

```
SOURCE: s=userSource(id("graphdataset"))
DATA: Category=col(source(s), name("Category"), unit.category())
DATA: Volume=col(source(s), name("Volume"))
COORD: transpose()
GUIDE: axis(dim(1), label("Fulfillment Defect Category"))
GUIDE: axis(dim(2), label("Incident Count"), null())
SCALE: linear(dim(2), include(0))
```

Graphics Production Language (GPL) describes how SPSS draws a graph: its data, axes, labels and marks. Read each block's purpose; you do not need to memorize it.

Check the chart's baseline, units, labels and category order. A neutral palette means restrained colors. A zero baseline preserves bar-length comparisons.

This is optional supplied code. For the required lab, use the delivered syntax and inspect its output. Writing GPL is not an additional task.

SPSS GPL DECLUTTERED BAR CHART SPECIFICATION

```
ELEMENT: interval(position(Category*Volume), color.interior(color.grey),  
transparency.interior(0.2))  
END GPL.
```

Graphics Production Language (GPL) describes how SPSS draws a graph: its data, axes, labels and marks. Read each block's purpose; you do not need to memorize it.

Check the chart's baseline, units, labels and category order. A neutral palette means restrained colors. A zero baseline preserves bar-length comparisons.

This is optional supplied code. For the required lab, use the delivered syntax and inspect its output. Writing GPL is not an additional task.



Group related information

GROUPING AND VISUAL ATTENTION

Use spacing and emphasis to guide the eye

1

Notice a cue

- **Sensory** vision notices some differences before deliberate attention. These preattentive cues include position, length and color. They guide the eye but do not guarantee correct interpretation.

2

Group related information

- **Gestalt** principles describe perceived grouping: proximity means nearby items seem related; similarity means matching appearance suggests a group; enclosure means a shared boundary connects items.

3

Apply the idea

- Place each value beside its hub label, align comparable figures and use one consistent accent for the selected hub. Keep labels close to their evidence. Leave more space between unrelated groups, and do not rely on color alone.



Equal medians can hide
unequal spread

INTERPRETING HUB DISTRIBUTIONS

Read center, spread and flagged observations together

1

Compare the middle

- All hubs have **median** 22.0 minutes. Hubs A and B each have **IQR** 7.0 minutes; Hub C has **IQR** 17.0 minutes. Equal medians can hide unequal variability.

2

Inspect the upper tail

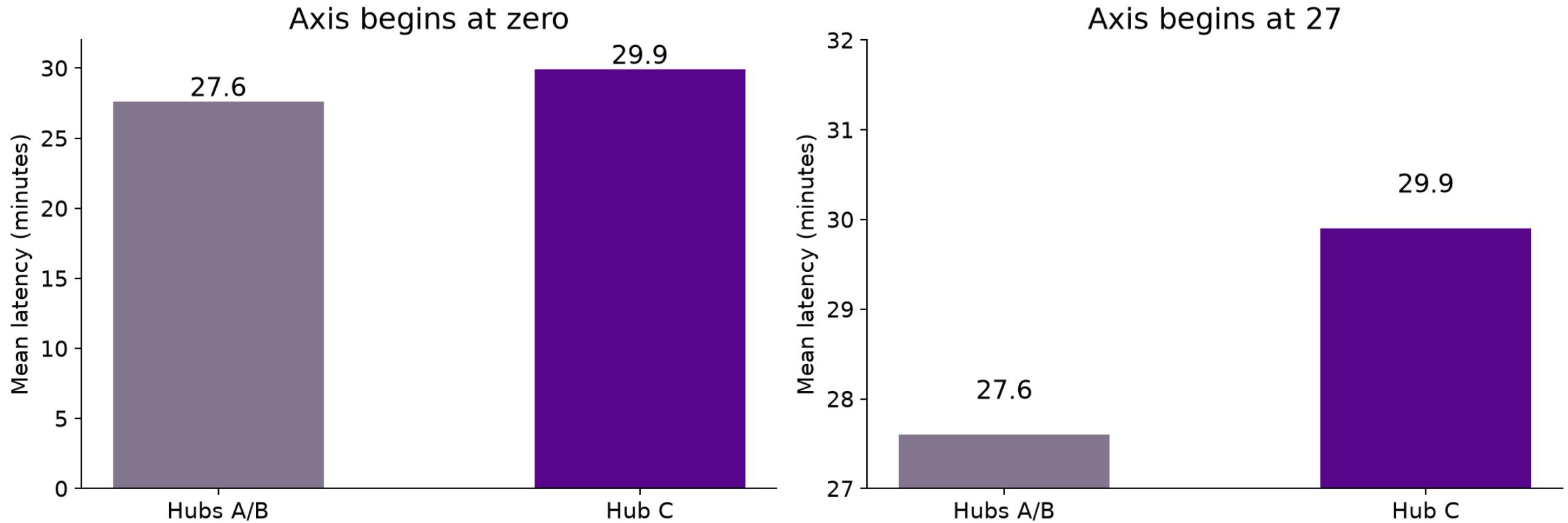
- Across all valid latencies, **IQR** is 12.0 minutes. The upper **Tukey** fence is 49.0 minutes and flags 829 records.
- The two largest valid latencies are both 124.0 minutes. A flag marks an unusual value for review; it is not automatic proof of error.

3

Keep the claim descriptive

- Hub C's middle half is more spread out. Investigate the process and service mix before attributing that difference to a carrier.

TWO DISPLAYS, ONE DATASET



Both panels show A/B mean 27.6 and Hub C mean 29.9 minutes. Compare how the axis changes your impression.

DECISION POINT: WHICH DISPLAY EARNS TRUST?

Use the two charts on the preceding slide.



New evidence

Both charts show the same means: 27.6 and 29.9 minutes. One bar chart starts at zero; the other at 27. Presentation changes the apparent size of the gap.

A

Apply the penalty now

Benefit: immediate accountability.

Risk: blaming the carrier for the wrong cause.

B

Hold one cycle for an audit

Benefit: stronger evidence of the cause.

Cost: Metro funds support during the delay.

Your decision

Reconsider A/B using the values. Which display would you bring to management, and why?



Excel first, SPSS second

PRACTICE C: CALCULATE, CHECK, SAVE

Follow the same Part in the lab guide

1

Open

- IN-CLASS_Lab Part C in your saved workbook. The lab guide keeps the formula and explanation together.

2

Do now

- **6 minutes:** complete **B7:B11**. Use the feedback beside the row, then save.

3

Next

- Interpretation and **SPSS H3** belong to the Section 4 evidence clinic.

SECTION

GROUPED SUMMARIES, COMPARATIVE DISTRIBUTIONS, AND LAB PRACTICE

Group comparable records and summarize each group. Keep data, calculations and explanations traceable so another analyst can repeat your work. This makes the analysis reproducible.

Grouped Tables · Reshaping Data · Lab Practice

STACKED DATA: ONE OBSERVATION PER ROW

Triage groups patients by urgency; these labels are illustrative.

Patient_ID	Triage_Level	Wait_Minutes
PT-101	Urgent	12
PT-102	Priority	28
PT-103	Routine	45
PT-104	Priority	34
PT-105	Urgent	18
PT-106	Routine	52

Stacked (long) data keeps each observation on a row. Patient_ID identifies the patient; Triage_Level gives the urgency group; Wait_Minutes records the wait.

Six-patient introductory excerpt of triage intake | Morning shift triage intake excerpt | Unit: Minutes waited



Keep groups and values
explicit

STACKED AND UNSTACKED DATA

Choose a layout that supports the comparison

1

Stacked, also called long

- Each observation has a row. A group column identifies Urgent, Priority or Routine; a value column records Wait_Minutes. The previous slide shows this layout.

2

Unstacked, also called wide

- Separate columns hold each group's values. Unequal group sizes can leave empty spaces; these spaces do not necessarily mean a patient's wait is missing.

3

Match the layout to the task

- A PivotTable is an Excel table that groups rows and calculates summaries. Stacked data supports the grouped PivotTable and **SPSS** procedures used here.
- Wide layouts also serve other tasks, such as paired before-and-after measurements. Keep row meaning and identifiers clear when reshaping.

SUBGROUP SAMPLE MEAN AND POOLED VARIANCE

$$\bar{x}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} x_{ij}, \quad s_j^2 = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2$$

Subgroup means one group within the full dataset. j labels a group; n_j is its valid count; x_{ij} is observation i in group j . $\bar{x}_j$ is that group's mean; s_j^2 is its sample variance.

Compute each group's summaries using its own observations and report the count.

Pooled variance combines group variances, weighted by $n_j - 1$, their degrees of freedom. It does not establish equal spread or comparable conditions.

Pooling is optional here. The required lab compares counts, medians, quartiles and IQRs.

SPSS MEANS AND GROUP BREAKDOWN SYNTAX

```
MEANS TABLES=Wait_Minutes BY Triage_Level  
  /CELLS=COUNT MEAN STDDEV MEDIAN.  
  
EXAMINE VARIABLES=Wait_Minutes BY Triage_Level  
  /PLOT=BOXPLOT  
  /STATISTICS=DESCRIPTIVES.
```

Transfer example: hospital waiting times. MEANS groups Wait_Minutes by Triage_Level; EXAMINE adds distribution summaries and plots.

Find each group's valid count, mean, median and spread in the Output Viewer. Distinguish a group label from the numeric response.

Different case mix, meaning different patient needs, can explain part of an observed difference. The output alone does not isolate a cause.

For the logistics lab, use the delivered Hub and Latency_Minutes commands.



Group, calculate, then
interpret

HOW A PIVOTTABLE GROUPS RECORDS

Summarize the observations within each category

1

Select the source

- Use a table with one observation per row, clear headers and a separate group field. Check missing values and duplicates before summarizing.

2

Choose groups and summaries

- Insert a PivotTable. Put Triage_Level in Rows and Wait_Minutes in Values. Use Value Field Settings to select **Count**, **Average** or **StdDev** (sample standard deviation) as needed.

3

Read what was grouped

- For Urgent waits of 12, 18 and 15 minutes, **Count** = 3 and **Average** = 15 minutes. The **count** is the number of valid waits.
- A standard PivotTable does not offer **median** or quartiles. Use the supplied formulas for those summaries.



Define, screen, summarize
and explain

A DESCRIPTIVE AUDIT

Build a trustworthy baseline before recommending action

1

Define and screen

- State the business question, population (the people or events of interest), reporting period and row meaning. Check keys, missing-value codes and units. Record every exclusion rule.

2

Summarize and compare

- For each relevant group, calculate the valid count, **mean**, **median** and spread. Inspect the distribution with a suitable chart using the same valid observations.

3

Report evidence and limits

- Describe what differs, how large the difference is and what remains uncertain. A group difference starts an investigation; it does not prove a cause.
- Keep raw data, calculations and explanations traceable so another analyst can repeat the result.

EXCEL DYNAMIC GROUPED DESCRIPTIVE SUMMARY AND AUDIT REFERENCE

```
=AVERAGEIFS(Daily_Sales, Store_Format, "Flagship")  
=MEDIAN(IF(Store_Format="Flagship", Daily_Sales))  
=STDEV.S(IF(Store_Format="Flagship", Daily_Sales))  
=QUARTILE.EXC(IF(Store_Format="Flagship", Daily_Sales), 3) -  
QUARTILE.EXC(IF(Store_Format="Flagship", Daily_Sales), 1)
```

Transfer example: store sales. AVERAGEIFS selects a named category; IF selects the matching observations for median, SD and quartiles.

Check that every statistic uses the same group and valid-record rule. Q3 minus Q1 gives the group's IQR.

The example's sales figures are not the logistics lab's expected values.

For LAB01 Part D, use the supplied formulas referencing Orders[Hub] and Orders[Latency_Minutes], then check that the counts agree.

SPSS GROUPED DESCRIPTIVE BREAKDOWN VIA MEANS AND EXAMINE SYNTAX

```
SORT CASES BY Store_Format.  
SPLIT FILE LAYERED BY Store_Format.  
DESCRIPTIVES VARIABLES=Daily_Sales  
  /STATISTICS=MEAN STDDEV MIN MAX.  
EXAMINE VARIABLES=Daily_Sales BY Store_Format  
  /PLOT BOXPLOT HISTOGRAM  
  /STATISTICS DESCRIPTIVES
```

Transfer example: store formats. SORT CASES orders the grouping variable; SPLIT FILE requests separate output by group; DESCRIPTIVES and EXAMINE produce the summaries.

Turn SPLIT FILE OFF afterward so later results are not accidentally grouped.

Locate the group label, valid count and units before comparing tables.

Use the delivered LAB01 H4 block for the required hub comparison. A difference in spread remains descriptive evidence.

SPSS GROUPED DESCRIPTIVE BREAKDOWN VIA MEANS AND EXAMINE SYNTAX

```
/PERCENTILES(25,50,75).  
SPLIT FILE OFF.
```

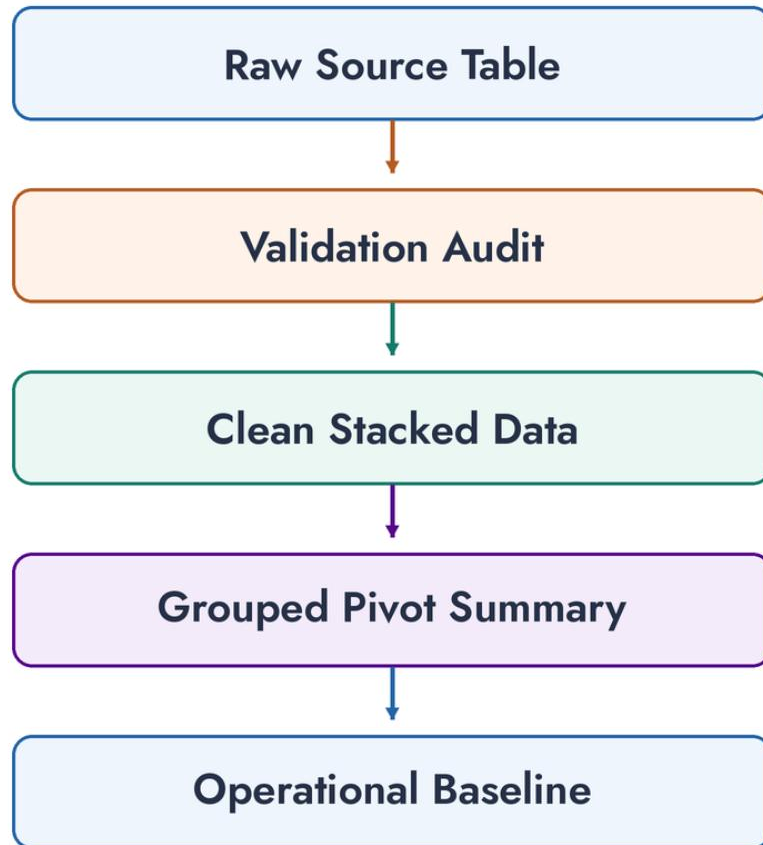
Transfer example: store formats. SORT CASES orders the grouping variable; SPLIT FILE requests separate output by group; DESCRIPTIVES and EXAMINE produce the summaries.

Turn SPLIT FILE OFF afterward so later results are not accidentally grouped.

Locate the group label, valid count and units before comparing tables.

Use the delivered LAB01 H4 block for the required hub comparison. A difference in spread remains descriptive evidence.

STRUCTURING REPRODUCIBLE ANALYSIS WORKBOOKS



Keep the source intact

Preserve the original data. In this lab, that is the Data sheet; do not overwrite sentinels or identifiers.

Make cleaning visible

Use formulas that explicitly exclude missing values. A blank is not a measured zero.

Make results traceable

Keep each summary beside its explanation. Another person should be able to follow the result back to the source and repeat it.

AI SEARCH & SHARE



Discuss with a partner; AI support is optional.

APPLY THE IDEA

Discuss with a partner; AI support is optional.

1

Consider

- What is the primary operational risk of establishing a single chain-wide performance target based on the combined mean across all retail branches?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: WHAT CAN WE ATTRIBUTE?

Attribution means assigning an outcome to a cause.



New evidence

The data compares hubs, but does not control route mix, demand, weather or carrier handoffs. Wider spread identifies a service concern; it does not establish who caused it.

A

Apply the penalty now

Benefit: immediate accountability.

Risk: blaming the carrier for the wrong cause.

B

Hold one cycle for an audit

Benefit: stronger evidence of the cause.

Cost: Metro funds support during the delay.

Your decision

Choose A/B. Name one missing fact and an interim customer protection.



Excel first, SPSS second

PRACTICE D: CALCULATE, CHECK, SAVE

Follow the same Part in the lab guide

1

Open

- IN-CLASS_Lab Part D in your saved workbook. The lab guide keeps the formula and explanation together.

2

Do now

- **7 minutes:** complete **B7:F10**. Use the feedback beside the row, then save.

3

Next

- Chart interpretation and **SPSS H4** begin the existing take-home work.



LAB

12 minutes · return to your saved work



Screen, summarize,
visualize, and compare
fulfillment data.

USE YOUR RESULTS TO MAKE A CLAIM

Follow the interpretation and evidence rows in each Part

1

First 4 minutes: Part A

- Explain the sentinel checks and the scale choice. Run **H1** and record the requested evidence in the Part A response cells.

2

Next 4 minutes: Part B

- Explain mean versus median and **SD** versus **IQR**. Run **H2** and compare its output with your checked Excel results.

3

Final 4 minutes: Part C

- Explain what the fences flag. Run **H3**, caption the output, and save. Resume Part D interpretation in the take-home work.



Listen and explain your reasoning.

METRO NEEDS A NEXT STEP

Read the situation, then hear your group's views

1

The situation

- Customers are reporting delays. Hub C has the same median wait as the other hubs, but its delivery times are less consistent.

2

The choice

- Metro can apply a carrier penalty now (A), or wait one reporting cycle and investigate first (B). Customer support continues either way.

3

Your team's job

- Recommend an action and consider its consequences. Then apply today's evidence to the second question on your group page.



Listen and explain your reasoning.

TWO QUESTIONS, ONE GROUP RESPONSE

20 minutes total · discuss both questions together

1

Question 1 · Reflect on the decision

- Recommend A or B. Explain your reason and one unintended harm that your choice could cause. Different choices can be reasonable.

2

Question 2 · Apply the evidence

- Use the question written for your group. Explain a checked result or an important condition for using it. No new calculations are required.

3

Make time to listen

- Use 3 minutes to read and choose roles; 10 to discuss and record; 7 for class sharing. Within discussion: about 5 minutes for Question 1, 4 for Question 2, 1 for participation.



Listen and explain your reasoning.

DISCUSS BEFORE YOU WRITE

Use the context on your group page to begin

1

Hear each person

- Read the question aloud and pause to think. Invite each person to offer an idea, ask for clarification, or name a concern.

2

Write two short responses

- Aim for 40–70 words per question; maximum 90. Explain the group's reasoning. You may record a majority recommendation and a different concern.

3

Record participation

- Each member records their role, specific contribution, and in-person attendance. Prepare one insight from Question 2 for your reporter to share.



Listen and explain your reasoning.

SHARE ONE USEFUL INSIGHT

Each group has about 45 seconds to share its Question 2 response

1

Explain what matters

- Share your main point and the evidence or condition behind it. State a limit that affects how confidently you can use it.

2

Listen across groups

- Notice how the other groups' perspectives affect your first **A/B** choice. You may keep your recommendation or change it with a reason.

3

Submit your group's work

- Review the two responses and participation record together. Follow the submission instructions on the final page of the group handout.



Save, resume, reflect

CONTINUE FROM YOUR CHECKPOINT

Finish the same investigation

1

Save

- Keep your personalized workbook and **SPSS** files.

2

Resume

- **Start with Part D:** chart, **B12**, **B13:H15**, and **H4** evidence in **B17:H18**. Then continue at POST-CLASS_Part 1, **A7**.

3

Finish

- Use the existing 60-minute take-home period. Check your evidence and follow the issued submission requirements.

DECISION POINT: WHAT CHANGED YOUR MIND?

Return to your opening vote

A

Apply the penalty now

Explain why the contract allows a penalty and what evidence links the carrier to the delays. Name the risk of acting too soon.

B

Hold one cycle for an audit

Explain why comparable records need checking and how to protect customers while waiting. Name the cost of delay.

Your final position

Vote again. Use one verified result, explain what changed, and name a condition that could reverse your recommendation.



Carry forward the habit of auditing evidence before acting on a KPI.

THANK YOU

Carry forward the habit of auditing evidence before acting on a KPI.

1

Meaning for the decision

- You can now screen operational data, select admissible summaries, interpret center and spread under skewness, construct honest displays, and compare groups without overstating causality.

2

Optional question, not homework

- What additional evidence would Metro need to judge whether changing its service-level agreement (**SLA**), the contract's promised service standard, could reduce Hub C's extreme delays?



Distributed course materials and software documentation support today's methods.

SESSION REFERENCES

Distributed course materials and software documentation support today's methods.

1

Meaning for the decision

- The session draws on the distributed companions for **S03.U01** through **S03.U08**, including measurement scales, screening, center, dispersion, visual choice, decluttering, stacked data, and descriptive audit practice.
- Microsoft Excel 365 documentation supports the descriptive functions, PivotTable workflow, and chart tools used in the lab.

2

Meaning for the decision

- **IBM SPSS** Statistics documentation supports the **MEANS**, **EXAMINE**, and grouped descriptive procedures demonstrated in the session.

APPENDIX

APPENDIX A: SELECTED STUDY SUMMARIES

Review the examples, then explore the Session Study Toolkit.

How to use this appendix

Follow each example through its inputs, method and interpretation. Find complete explanations and references under the matching topic in the Session Study Toolkit.



Classify before calculating

WORKED AUDIT: MATCH FIELDS TO SUMMARIES

E-commerce service records: classify before calculating

1

Labels and ranks

- Customer IDs 10421, 10422 and 10423 identify records. **Count** or check uniqueness; an **average ID** has no useful meaning.
- Order Tier 1–3 ranks service classes. Report category frequencies or the middle category without assuming equal gaps.

2

Measured quantities

- Call durations of 124–680 seconds and invoices of \$18.50–\$240.00 measure quantities. State units before calculating a **mean**, **median** or total.

3

Choose a defensible claim

- **Return Status is Yes/No:** report counts or proportions. Satisfaction 1–5 gives ordered ratings; inspect category frequencies and the middle rating.
- A numerical format does not by itself justify arithmetic.



Check records and fields separately

WORKED AUDIT: SCREEN EACH FIELD

Separate retail example: seven raw transaction rows

1

Restore the intended row meaning

- One row should represent one order. Two rows repeat Order_ID 1004 with identical values. Verify that importing the data repeated a row, retain one and document the change.

2

Identify missing values

- The example defines age -1 and 999 as missing codes. A blank age is also missing. A blank delivery duration belongs to an order still in transit.

3

Use the right denominator

- After removing the verified duplicate, five completed durations remain: 3, 4, 2, 5 and 3 days. **Mean** = $17 \div 5 = 3.4$ days.
- Retain a valid duration even when age is missing. A blank duration is not zero days.



Define the population
before filtering

WORKED AUDIT: DEFINE WHICH DAYS COUNT

Separate retail study example: daily store revenue

1

Define the question

- The retail network compares revenue on operating days across four store formats. Each row is one store-day, with a format and Daily_Sales in dollars.

2

Apply the stated rule

- There are 233 records. Four zero-sales days were documented maintenance closures. Excluding these leaves 229 operating days for this specific comparison.

3

Explain the consequence

- If the question were total revenue across every calendar day, the closures would still matter. A genuine zero is not a missing value.
- Report group counts, center and spread with the exclusion rule. A higher **mean** does not prove that a format's marketing caused higher sales.



A typical payout and total payroll

WORKED EXAMPLE: MEAN AND MEDIAN PAYOUT

Ten sales representatives: commission is pay linked to sales; amounts are dollars

1

Keep all ten observations visible

- \$3,200; \$3,400; \$3,600; \$3,900; \$4,200; \$4,500; \$4,800; \$5,100; \$7,500; \$35,000. The final payout includes a large enterprise deal.

2

Calculate two centers

- **Median** = **average** of positions 5 and 6 = $(\$4,200 + \$4,500) \div 2 = \$4,350$.
- **Mean** = $\$75,200 \div 10 = \$7,520$. The large final payout pulls the **mean** above most individual payouts.

3

Match the result to the question

- The **median** describes the middle payout. The **mean** multiplied by 10 recovers the total cash needed. Neither number alone describes the full distribution.



Use every observation in
sample spread

WORKED EXAMPLE: CALCULATE SAMPLE SPREAD

Ten shipment durations, measured in minutes

1

Start with all ten durations

- 22, 24, 25, 26, 28, 29, 31, 33, 48, 74. Their sum is 340; **mean** = 34. **Median** = $(28 + 29) \div 2 = 28.5$.

2

Square each distance from 34

- First five:** $144 + 100 + 81 + 64 + 36 = 425 \text{ min}^2$. Remaining five: $25 + 9 + 1 + 196 + 1,600 = 1,831 \text{ min}^2$.

3

Finish and interpret

- Sample variance = $(425 + 1,831) \div 9 = 250.67 \text{ min}^2$. Sample **SD** = $\sqrt{250.67} = 15.83 \text{ minutes}$.
- The 74-minute shipment contributes $1,600 \text{ min}^2$. Squaring distances gives unusually distant observations strong influence.



A flag is a reason to investigate

WORKED EXAMPLE: INVESTIGATE HUB C'S TAIL

Metro latency: use the lab's specified quartile method

1

Read the middle half

- Hub C has **Q1** = 14, **median** = 22 and **Q3** = 31 minutes. **IQR** = $31 - 14 = 17$ minutes.

2

Calculate the upper fence

- Upper fence = **Q3** + $1.5 \times \text{IQR}$ = $31 + 25.5 = 56.5$ minutes. A 124-minute duration exceeds this boundary.

3

Interpret the flag

- The fence identifies observations to investigate. Check the original record, timestamps and operating conditions before deciding whether correction is justified.
- A valid long delay is part of customer experience. Keep it visible and report the distribution; do not delete it simply to improve the summary.



Read group size, center
and spread

WORKED EXAMPLE: COMPARE TRIAGE WAITS

Eight patients: an illustrative morning intake

1

Group the observations

- **Urgent:** 12, 18, 15 minutes. Priority: 28, 34, 31. Routine: 45, 52. Store one patient per row with Triage_Level and Wait_Minutes.

2

Build and read the PivotTable

- Put Triage_Level in Rows. Add Wait_Minutes to Values three times; select **Count**, **Average** and **StdDev**.
- **Expected count / mean / sample SD:** Urgent 3 / 15 / 3; Priority 3 / 31 / 3; Routine 2 / 48.5 / 4.95. Durations are minutes.

3

Interpret cautiously

- Routine patients waited longer in this small sample. Group size, arrival patterns and patient needs matter; the table alone does not explain why.

APPENDIX B: EXPLORATION AND STRETCH

Optional independent study

Meaning for the decision

Explore median absolute deviation (MAD), a measure of spread based on distances from the median. Follow the worked example through the center, distances and screening flag.

RESISTANT SPREAD

$$\text{MAD} = \text{median}_i |x_i - \tilde{x}|$$

$$\text{MADN} = \text{MAD} / 0.6745$$

$$M_i = 0.6745(x_i - \tilde{x}) / \text{MAD}$$

Median absolute deviation (MAD) is the middle distance from the median, ignoring direction. Normalized MAD (MADN) rescales it for a normal, bell-shaped model.

For 12, 14, 15, 16, 18, 22, 110: median = 16. Distances sort to 0, 1, 2, 2, 4, 6, 94; MAD = 2 minutes; MADN = 2.97.

M_i is a modified z-score: a scaled distance from the median. Its magnitude for 110 is 31.70.

A large score flags an observation to investigate. Tied (repeated) values can make MAD zero.

MAD WORKED EXAMPLE

Seven hypothetical pack durations

Duration	Absolute deviation
12	4
14	2
15	1
16	0
18	2
22	6
110	94

Absolute deviation means distance from the 16-minute median, ignoring direction. The 110-minute duration is 94 minutes away; the middle distance is only 2 minutes.

Seven hypothetical warehouse pack durations | Single teaching sample | Unit: minutes



Find the resistant center

MAD STEP 1: FIND THE MEDIAN

Seven illustrative pack durations, in minutes

1

Order the observations

- 12, 14, 15, 16, 18, 22, 110. There are seven values, so the middle position is $(7 + 1) \div 2 = 4$.

2

Use the resistant center

- The fourth value is 16 minutes. Three observations lie on each side. Increasing 110 further would not change this **median**.

3

Prepare to measure spread

- **Median** absolute deviation (**MAD**) measures the typical absolute distance from the **median**. First calculate each distance from 16; then find the middle distance.
- For example, 12 is 4 minutes away and 110 is 94 minutes away. Both distances are positive.



Order the distances, then
find the middle

MAD STEP 2: FIND THE MIDDLE DISTANCE

Use the distances from the 16-minute median

1

Sort the absolute distances

- The distances are 4, 2, 1, 0, 2, 6 and 94 minutes. Sorting gives 0, 1, 2, 2, 4, 6, 94.

2

Calculate MAD

- The fourth distance is 2 minutes, so **MAD** = 2 minutes. It describes distance from the **median**; it is not the **median** duration itself.

3

Understand the scaled version

- Normalized **MAD** (**MADN**) = **MAD** ÷ 0.6745 $\approx$ 2.97 minutes. This scaling puts **MAD** on a standard-deviation-like scale for normally distributed data.
- The scaling does not make the observations normal or guarantee equality with sample **standard deviation**.



Investigate before
excluding

MAD STEP 3: INVESTIGATE AN EXTREME

A screening flag starts a review

1

Compare two measures of spread

- For these seven durations, sample **SD** is **35.61** minutes; **MADN** is about 2.97 minutes. The 110-minute observation strongly influences **SD**.

2

Calculate the modified score

- The modified z-score is a scaled distance from the median. For 110 minutes: $0.6745 \times (110 - 16) \div 2 = \mathbf{31.70}$. An absolute score above **3.5** is a common potential-outlier flag.

3

Decide using evidence

- Check units, timestamps and process records. Correct a verified recording error; retain a genuine long event and explain its effect.
- If **MAD** is zero, this score cannot be calculated by division. Use other evidence and a suitable screening approach.

SECTION

APPENDIX C

Centers for Medicare & Medicaid Services (CMS): hospital visit measures

Optional post-session reading outside the live timetable.



Arriving for emergency care [Illustrative image]

CMS OP-18: WHAT IS MEASURED?

Emergency-department visit duration

1

The service question

- The Centers for Medicare & Medicaid Services (**CMS**), a **U.S.** health agency, reports emergency-department (**ED**) visit duration through outpatient measure **OP-18**.

2

From arrival to departure

- For each eligible visit, **elapsed minutes** = departure time – arrival time. Sort those durations; the **median** is the middle value.
- This covers the **whole visit**, including care and waiting. It is not simply time until a clinician first sees the patient.

3

Name the patient group

- **OP-18a** includes patients sent to another facility (transfers); **OP-18b** excludes transfer and psychiatric/mental-health patients. **OP-18d** describes transfers separately. Compare the same measure variant and reporting period.



Completing an emergency-department visit [Illustrative image]

WHY USE THE MEDIAN?

A typical visit and unusually long visits

1

Read the middle

- In the five-visit example on the next slide, a **median** of **165 minutes** means two visits were shorter and two were longer. Individual durations still varied.

2

Understand resistance

- The **median** depends on order, so one very long visit has less influence than it has on the **mean**. **Resistant** means less sensitive, not unaffected.
- The **mean** uses every duration. A large gap between **mean** and **median** is a reason to inspect the distribution.

3

Keep the long visits visible

- The **median** alone hides spread and the longest experiences. Also examine the range, upper percentiles and patient groups before judging service performance.



Reviewing eligible visit records [Illustrative image]

WHICH VISITS COUNT?

Worked teaching example: an OP-18b-style group

1

Define the group first

- **Illustrative records, not CMS results:** five non-psychiatric visits discharged home lasted 145, 155, 165, 175 and 720 minutes. A transferred visit lasted 210 minutes.
- Assume the five home discharges meet all other eligibility rules. This example excludes transfers; **OP-18a** would include them.

2

Calculate with the five eligible visits

- **Median** = third ordered value = **165 minutes**.
- **Mean** = $(145 + 155 + 165 + 175 + 720) \div 5 = \mathbf{272 \text{ minutes}}$. The long visit pulls the **mean** upward.

3

Explain the exclusion

- Exclude 210 because of the chosen **population rule**. Retain 720 if valid: being unusually long does not make a record wrong.



Discussing
service-improvement
evidence [Illustrative
image]

REPORT THE RESULT RESPONSIBLY

Make the number understandable and comparable

1

Give the number a clear label

- Report the **median**, minutes, eligible visit count, patient group and reporting period together. Name the **OP-18** variant and the measurement rules used.

2

Show what the center misses

- A lower **median** can coexist with very long visits. Examine upper percentiles and subgroup patterns; check missing timestamps before comparing hospitals or periods.
- **For our five-visit example: median 165 minutes, mean 272 minutes**, range 145–720 minutes. Each summary answers a different question.

3

State what the result cannot prove

- A shorter visit does not by itself prove better clinical care or explain why performance changed. Use **additional evidence** before making a causal claim.